Resolving Coordination Frictions in Green Labor Transitions:
Minimizing Unemployment, Costs, and Welfare Distortions

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ABSTRACT

Achieving ambitious decarbonization targets requires rapid reallocation of labor into green production but is slowed by a two-sided coordination failure: workers face entry costs and uncertain prospects, while firms underinvest without a reliable talent pool. I develop a two-sector Diamond–Mortensen–Pissarides model with worker entry costs, vacancy creation, and an output-equivalent environmental externality to compare worker-side, firm-side, and coordinated dual policies on target attainment, unemployment, fiscal cost, and welfare. Calibrated to a U.S. transition from 2% to 14% green employment by 2030 ($\approx$24 million jobs), one-sided policies reach the target but with unnecessarily higher unemployment and fiscal costs. A dual policy attains the same 14% share with 0.39 percentage points lower unemployment and 0.05 percentage points of GDP lower fiscal outlays than the cheapest one-sided alternative, while minimizing welfare losses. Optimal policy is dynamic: total support is hump-shaped over the adoption path, and the composition rotates: worker support is most valuable early and late, while firm-side support is most effective mid-transition. An output-equivalent environmental externality of about 0.88% of GDP is sufficient for the transition to raise aggregate welfare. The core lesson is that policy design, not just policy level, matters: treating workers and firms as co-dependent levers and adjusting the mix over time yields a socially inclusive and fiscally sustainable transition.

This Version: October 23, 2025

JEL Classification: E24, E60, J64, Q54, Q58, O44
Keywords: search and matching, unemployment, labor market entrants, green transition, climate policy

I am grateful to Thanasis Geromichalos for guidance and comments, and to Si Guo, Emile Marin, Katheryn Russ, Oscar Jorda, Nicolas Karamp, Ioannis Kospentaris, Sukjon Lee, Lucas Herrenbrueck for helpful discussions. I also thank seminar participants at UC Davis, the Center for Analytical Finance (CAFIN) at UC Santa Cruz, the IMF Climate Analytics Hour, and the California Council on Science and Technology (CCST) Showcase for valuable feedback. All remaining errors are my own.

1 Introduction

Managing large structural shifts, such as decarbonization, digitalization, and trade realignment, has become a central challenge for policymakers. These transitions create opportunities for long-term growth but also generate costly distortions when labor reallocation is slow or one-sided. The green transition is no exception: moving workers out of carbon-intensive activities and into clean production will not proceed frictionlessly. Left unchecked, the process can raise unemployment, strain public budgets, reduce aggregate welfare, and delay progress toward carbon neutrality by 2050.

This paper starts by diagnosing why markets underdeliver during such transitions: large reallocations are slowed by a two-sided coordination failure. Workers face training and mobility costs and hesitate to switch sectors when job prospects are uncertain. Firms hesitate to create vacancies when the supply of qualified workers is uncertain. Each side waits for the other, resulting in under-entry and under-vacancy creation, and hence stalling the intended transition. Crucially, the misalignment is structural: workers bear the up-front entry cost, while firms realize any production subsidy only once a match forms and output is produced. Evidence from green labor markets supports this chicken-and-egg problem: workers in carbon-intensive activities face rising unemployment fears and uncertainty, while green job employers report hiring bottlenecks; postings for green jobs have grown nearly twice as fast as the available workforce, with multi-million-worker shortfalls projected by 2030 (Bluedorn, Hansen, Noureldin, Shibata, and Tavares, 2023; LinkedIn Economic Graph, 2024; ILO, 2019; BCG, 2023). In search environments, such two-sided externalities imply laissez-faire under-reallocation and a clear role for policy (Diamond, 1982; Cooper and John, 1988; Mortensen and Pissarides, 1994; Pissarides, 2000). The purpose of the paper is to build on this coordination friction and study how policy should be designed.

Given the goal of expanding the green sector, an objective that can be justified by environmental externalities, the question is not whether to intervene but how to do so. Among the many feasible ways to induce the transition, which policy mix attains the target most efficiently? I focus on three margins central to macro policy: the unemployment cost borne during reallocation, the fiscal resources required to deliver it, and the welfare consequences once environmental benefits are accounted for. The analysis then asks a dynamic question that existing work rarely tackles: how should the size and composition of support evolve along the path to the target? Methodologically, rather than solving a perfect-foresight transition path, I compute the path as a sequence of steady states that hit a schedule of intermediate adoption targets. This mirrors how policy is actually made via staged goals and budget cycles—interim targets are set, instruments are recalibrated, and budgets are updated at each waypoint—and it makes the instrument rules transparent while preserving discipline from the structural model.[1] The answers deliver implementable design rules that generalize to other large structural shifts with two-sided reallocation constraints.

I develop a two-sector Diamond–Mortensen–Pissarides framework in which workers can enter a growing green market by paying an entry cost, and firms create vacancies subject to posting costs. The government has two instruments: a level subsidy to green entrants and a per-unit production subsidy to green firms, both financed under a balanced budget. The worker subsidy is paid up front upon entry, whereas the firm-side production subsidy is realized only when a match forms and output is produced. The quantitative illustration calibrates the model to U.S. labor-market moments and sectoral shares and targets an increase in green employment from about 2 percent to 14 percent by 2030, nearly 24 million green jobs, consistent with widely cited roadmaps to be on track to net-zero (WorkingNation, 2024). The scale anchors the exercise in a realistic policy problem while the analysis highlights general mechanisms.

Which side of the market the government chooses to assist matters for equilibrium outcomes. One-sided designs, worker-only or firm-only, can reach a given employment target but do so by leaving the other side as the bottleneck, thereby raising unemployment and the funding requirement during the transition. A coordinated dual policy that lowers worker entry costs while incentivizing green firms' production aligns both sides and dominates on the two margins that sit at the heart of green transition debates: jobs and fiscal sustainability. In the benchmark exercise, the dual design reaches the 14 percent target with 39 basis points lower unemployment and 5 basis points of GDP less fiscal outlay than the cheapest one-sided alternative. This static dominance is in line with insights from two-sided markets (Rochet and Tirole, 2003).

The paper then develops a dynamic design rule for two-sided transitions. Optimal policy is state-contingent in both intensity and composition. First, total support is hump-shaped over the adoption path: support is modest early on, peaks at intermediate adoption when neither side responds strongly, and then eases as the target share is approached. Second, the composition of support rotates with scale: worker support does more of the work early and late, when expanding or replenishing the entrant pool is decisive, while

[1]See UNFCCC, *The Paris Agreement* and UNFCCC, *Nationally Determined Contributions (NDCs)*: Parties submit NDCs on a five-year cycle and each successive NDC must represent a "progression" in ambition. See also International Energy Agency, *Net Zero by 2050: A Roadmap for the Global Energy Sector* (2021) and the *Net Zero Roadmap* (2023 update), which set out 400+ dated milestones (many by 2030/2035) rather than a single 2050 endpoint.

firm-side support is more critical in the middle, when vacancies must be filled through a temporarily tight hiring environment. These qualitative patterns follow from the forces of two-sided search with costly worker entry, vacancy creation by firms, and a balanced-budget instrument set. Early in the transition, it is relatively cheap to move marginal workers, and firms can absorb them. At intermediate scale, both margins become less responsive, pushing up the marginal fiscal cost. Late in the transition, even though the green sector has reached scale, meeting rates alone cannot bring these workers across as the residual pool of potential movers is small; the binding constraint reverts to new entry rather than matching from the existing pool. The dynamic composition therefore follows a worker–firm–worker sequence along the adoption path. The quantitative results confirm the magnitude and policy relevance of these forces.

The framework also speaks directly to the welfare question that motivates public action: when is expanding the green sector socially desirable? By incorporating environmental benefits as an output-equivalent externality, the model delivers a transparent threshold: once the social benefit of green matches exceeds a modest level, coordinated policy not only meets quantitative targets but also raises aggregate welfare relative to no intervention. The calibration suggests this threshold is equivalent to a 0.88% increase in output, well within ranges discussed in applied work.

This paper connects and extends several literatures. Macroeconomic models of decarbonization typically track sectoral reallocation but often assume frictionless labor mobility or full employment, overlooking issues like unemployment, search costs, and costly worker switching (Bilal and Stock, 2025; Patuelli, Nijkamp, and Pels, 2005; Sancho, 2010; Böhringer, Rivers, and Rutherford, 2013; Freire-González, 2018). Directed technical change literature emphasizes how policy steers innovation (Acemoglu, Aghion, Bursztyn, and Hemous, 2012; Popp, 2020), but often abstracts from two-sided labor reallocation. Recent macro-labor approaches incorporate search frictions (Hafstead and Williams III, 2018; Hafstead, Williams III, and Chen, 2022), but rarely go into detail of policy instrument rules tailored to the reallocation path. Empirically, environmental regulation and carbon pricing create modest but heterogeneous displacement with slow adjustment (Walker, 2011, 2013; Shapiro and Metcalf, 2023; Conte, Desmet, and Rossi-Hansberg, 2022), and international evidence documents skill bottlenecks and hiring frictions (ILO, 2019; Bluedorn et al., 2023).

Within macro-labor, search-and-matching studies emphasize hiring frictions and skill constraints (Gibson and Heutel, 2023; Lankhuizen, Rojas-Romagosa, and van Ewijk, 2022), while evaluations of active labor-market policies document mixed effectiveness of training and hiring subsidies (Card, Kluve, and Weber, 2018; Brown, 2015; Kluve, Puerto, Robalino, Romero, Rother, Stöterau, Weidenkaff, and Witte, 2019). In parallel, the two-sided markets literature clarifies why single-sided interventions are dominated in static settings (Rochet and Tirole, 2003); the contribution of this paper is to move from the static dominance result to a dynamic policy design in a search environment with a balanced budget. Relative to the DMP tradition (Mortensen and Pissarides, 1994; Pissarides, 2000), the theoretical step is to model sectoral entry on the worker margin as a level cost and to study its interaction with vacancy creation under dual instruments and balanced financing.

This paper makes five contributions that follow directly from the two-sided coordination failure and the staged transition design. First, it centers the twin constraints of the green transition, unemployment risk and the fiscal cost of scale, in the policy design problem. Second, it offers a generalizable two-sided framework for large reallocations where the same "chicken-and-egg" friction prevails: workers delay entry without visible jobs, and firms delay investment without a ready workforce. The framework lets policymakers compare instruments (e.g., training grants vs. hiring subsidies; production credits vs. procurement or loan guarantees) and funding methods along a staged sequence of intermediate targets, not just a distant endpoint. The same logic applies to AI adoption and digital upskilling (certification pipelines vs. firm investment in AI complements), transport electrification and grid build-out (EV/charging rollout vs. technician training and relocation), and industrial and infrastructure expansions (capital investment support vs. education/apprenticeship subsidies). Third, it embeds environmental externalities in the design problem and provides a transparent, output-equivalent welfare criterion for when the transition raises aggregate welfare. Fourth, it derives implementable dynamic rules: total support is hump-shaped over the adoption path, and the instrument mix rotates as the binding margin shifts (worker–firm–worker). Fifth, it shows quantitatively that coordinated dual policy dominates one-sided alternatives at a fixed target by jointly lowering unemployment and fiscal outlays, with a calibrated U.S. application illustrating decision-usefulness.

The remainder of the paper presents the model (Section 2), policy instruments and calibration (Sections 3–4), quantitative results on unemployment, funding, and welfare along the path to the 14% target (Section 5), and implications for the scale, timing, and composition of support (Section 6).

2 Model

I begin with a static benchmark of a modified Diamond–Mortensen–Pissarides (DMP) model to characterize equilibrium in the presence of coordination frictions that arise during green reallocation. The static environment allows a clear analysis of how subsidies and costs shape outcomes in segmented labor markets. Building on these insights, I then develop a dynamic version that characterizes equilibrium over time, allowing us to study unemployment dynamics, fiscal requirements, and aggregate welfare–central concerns for current green transition policies.

2.1 Static Model: DMP Framework

The static model provides a tractable setting to study how workers and firms interact across segmented labor markets when policy instruments target each side. I focus on (i) multiplicity arising from coordination frictions, (ii) how subsidies and costs affect equilibrium, and (iii) efficiency conditions.

2.1.1 Physical Environment

I consider a static (one-period) version of the DMP model (Pissarides, 2000) with two sectors: green (g) and non-green (n). The labor force is normalized to 1, and both workers and firms decide optimally which market to enter. A continuum of unemployed workers receive an unemployment benefit z. Let $\pi \in [0, 1]$ denote the share of unemployed workers who enter the green market (thus $1 - \pi$ choose the non-green market). Firms choose in which sector to post vacancies.

Firms that enter either sector incur a recruiting cost c. Green firms receive a production subsidy τ_g, while non-green firms face a tax τ_n that finances the subsidies.[2] On the worker side, entry into the green sector requires paying a flow cost κ_g, partially offset by a per-period subsidy $s_g \geq 0$. The effective cost is $\kappa_g - s_g$. Entry into the non-green sector is costless. Within each sector $j \in \{g, n\}$, matches are produced by a constant-returns technology:

$$m_j(u_j, v_j) \;=\; \delta_j \, \frac{u_j \, v_j}{u_j + v_j},$$

where u_j and v_j denote unemployed workers and vacancies. For the static calibration I normalize $\delta_j = 1$ (levels rescale meeting probabilities without affecting the qualitative

[2] I also solved a version in which all sectors are taxed to fund green subsidies. Since this does not change the qualitative results and adds notation, I present the simpler case in which only non-green production is taxed.

results).[3]

This matching function is symmetric in (u_j, v_j), homogeneous of degree one, strictly increasing in each argument, and bounded above by $\min\{u_j, v_j\}$. The implied meeting probabilities are

$$\alpha_{wj} \;\equiv\; \frac{m_j}{u_j} \;=\; \delta_j \, \frac{v_j}{u_j + v_j}, \qquad \alpha_{fj} \;\equiv\; \frac{m_j}{v_j} \;=\; \delta_j \, \frac{u_j}{u_j + v_j}, \qquad j \in \{g, n\}. \tag{1}$$

With tightness $\theta_j \equiv v_j / u_j$, these reduce to

$$\alpha_{wj} = \delta_j \, \frac{\theta_j}{1 + \theta_j}, \qquad \alpha_{fj} = \delta_j \, \frac{1}{1 + \theta_j},$$

so that $\partial \alpha_{wj} / \partial \theta_j = \delta_j / (1 + \theta_j)^2 > 0$ and $\partial \alpha_{fj} / \partial \theta_j = -\delta_j / (1 + \theta_j)^2 < 0$.

After matches form, wages are set by Nash bargaining. Workers' bargaining power is denoted by $\eta \in (0, 1)$. All other primitives are identical across sectors.

This environment embeds the coordination problem salient in green labor markets. Firms benefit from the production subsidy τ_g but will open vacancies only if they expect sufficient worker entry; workers face entry cost $\kappa_g - s_g$ and will invest only if they expect enough vacancies. Entry decisions on both sides are therefore mutually reinforcing. As I show below, this strategic complementarity generates multiple equilibria.

2.1.2 Workers

Unemployed workers receive z. If they enter the green market, they pay $\kappa_g - s_g$. Expected payoffs in each sector are:

$$\begin{aligned} \text{Green:} \quad & -(\kappa_g - s_g) + \alpha_{wg} w_g + (1 - \alpha_{wg})\, z, \\ \text{Non-green:} \quad & \alpha_{wn} w_n + (1 - \alpha_{wn})\, z. \end{aligned}$$

[3] I also considered the more general family $m_j = \delta_j \left(\frac{u_j v_j}{u_j + v_j} \right)^{1-\rho} (u_j v_j)^{\rho}$ with $\rho \in [0, 1]$, which nests increasing returns when $\rho > 0$; full IRS results are available upon request. I adopt the harmonic-mean CRS case ($\rho = 0$) because it yields bounded, transparent meeting rates $f(\theta) = \delta\,\theta/(1+\theta)$ and $q(\theta) = \delta/(1+\theta)$ with well-behaved derivatives $f'(\theta) = \delta/(1+\theta)^2 > 0$, $q'(\theta) = -\delta/(1+\theta)^2 < 0$ for all θ. By contrast, a Cobb–Douglas form $m = \delta\, u^{1-\phi} v^{\phi}$ implies $f(\theta) = \delta\,\theta^{\phi}$, $q(\theta) = \delta\,\theta^{\phi-1}$ with constant log-elasticities ϕ and $\phi-1$ and level responses that can be implausibly steep/flat near $\theta \to 0, \infty$. The harmonic-mean specification preserves CRS, symmetry, and global bounds on meeting probabilities which will be later used for policy analysis.

In equilibrium, workers are indifferent across markets,

$$-(\kappa_g - s_g) + \frac{v_g}{\pi + v_g} w_g + \left(1 - \frac{v_g}{\pi + v_g}\right) z \;=\; \frac{v_n}{(1-\pi) + v_n} w_n + \left(1 - \frac{v_n}{(1-\pi) + v_n}\right) z. \quad (2)$$

2.1.3 Firms

Free entry implies firms post vacancies until expected profits equal the recruiting cost c:

$$\begin{aligned} \text{Green:} \quad & c = \alpha_{fg}\big(p + \tau_g - w_g\big), \\ \text{Non-green:} \quad & c = \alpha_{fn}\big(p - \tau_n - w_n\big), \end{aligned}$$

where p is worker productivity (common across sectors). Subsidies and taxes shift the surplus split between workers and firms.

2.2 Wage Determination (Nash Bargaining)

With worker bargaining power η, the static Nash solution yields

$$w_g = \eta\,(p + \tau_g) + (1-\eta)\,z, \qquad w_n = \eta\,(p - \tau_n) + (1-\eta)\,z.$$

Derivations are in Appendix A.

2.2.1 Equilibrium

An equilibrium consists of wages (w_g, w_n), vacancies (v_g, v_n), unemployment and employment masses (u_g, u_n, e_g, e_n), the worker entry share π, and policy instruments (τ_g, τ_n, s_g) such that: (i) workers' indifference holds; (ii) firm free-entry conditions hold; (iii) matching probabilities satisfy (1); and (iv) market sizes add up to the unit labor force.

Proposition 1. *The static economy admits multiple equilibria with following properties*[4]*:*

- $\exists$ *a corner equilibrium where* $\pi = 0$ *and there are no jobs/production in the economy is green.*
- $\exists$ *a corner equilibrium where* $\pi = 1$ *and all jobs/production in the economy is green.*

[4]In a different context centered on asset liquidity, (Geromichalos, Herrenbrueck, and Lee, 2023) investigates how agents select among various asset markets when faced with random liquidity demands, deriving a solution that mirrors the characterization presented here.

- *For CRS case with $\psi = 0$, $\lim_{\pi \to 0^+} G(\pi) > 0 > G(0)$ and $\lim_{\pi \to 1^-} G(\pi) < 0 < G(1)$, i.e. the corner equilibria are not robust to small trembles, but $\exists$ at least one robust interior equilibrium.*

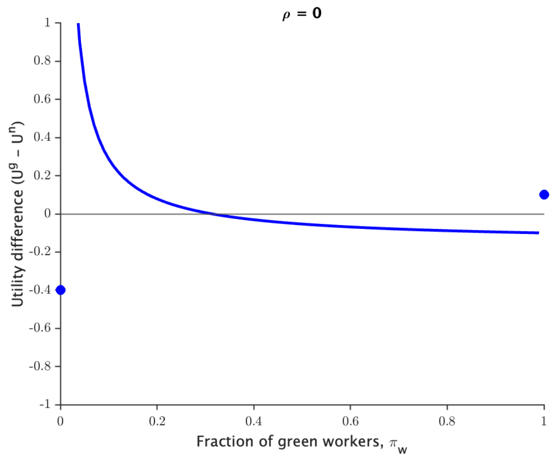
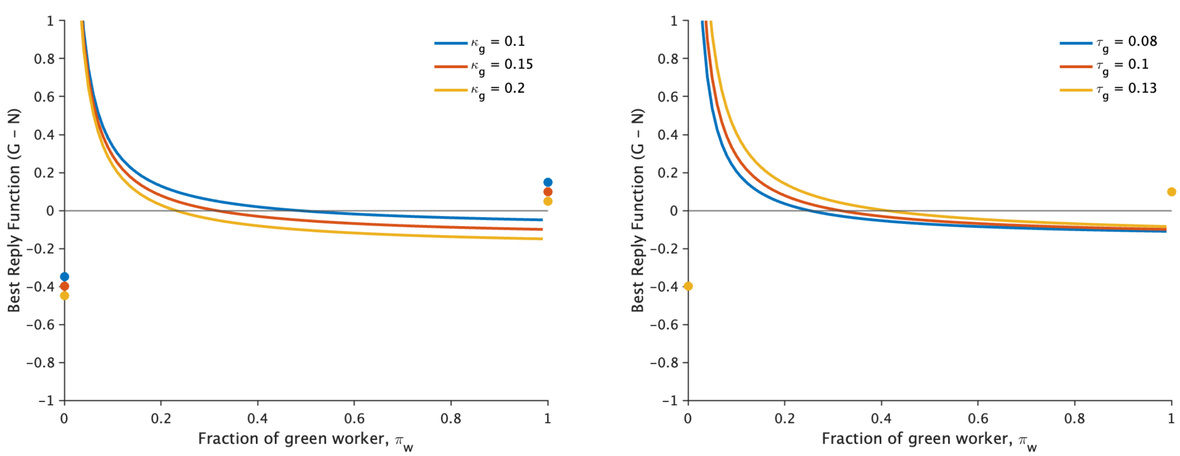


Figure 1. The reply function (G–N) for CRS ($\rho = 0$), showing multiplicity of equilibria in the static benchmark.

2.3 Comparative Statics

I conduct comparative statics in the static model to illustrate the main policy channels. Figure 2a shows that reducing workers' entry cost (higher s_g or lower κ_g) raises equilibrium green entry; Figure 2b shows that a higher firm subsidy τ_g likewise increases green entry.



(a) Higher worker cost reduces equilibrium green entry.

(b) Higher firm subsidy increases equilibrium green entry.

Figure 2. Comparative statics: entry costs and subsidies on equilibrium green entry.

Overall, the static benchmark highlights the need to incentivize *both* sides of the market. It also provides a clear framework for the efficiency analysis below, where I show how environmental externalities interact with the standard search externality and how policy instruments can decentralize the planner's allocation.

2.4 Social Efficiency and Policy Implications[5]

The decentralized equilibrium may be inefficient for two reasons: (i) standard search-and-matching externalities, and (ii) uninternalized environmental externalities from green job creation. I characterize efficiency using a social planner who internalizes both.

2.4.1 Planner's Problem Without Environmental Externalities

If green jobs yield no external benefits ($\phi = 0$), the planner maximizes total surplus:

$$\max_{\{\pi, v_g, v_n\}} L(\pi, v_g, v_n) = \underbrace{\pi \frac{v_g}{\pi + v_g}(p - z)}_{\text{Green matches}} + \underbrace{(1-\pi)\frac{v_n}{(1-\pi) + v_n}(p - z)}_{\text{Non-green matches}} - c(v_g + v_n) - \pi \kappa_g. \quad (3)$$

Proposition 2. *Without externalities ($\phi = 0$), the planner's solution is a corner: if $\kappa_g > 0$, then $\pi^* = 0$ and all workers enter the non-green sector. No interior allocation is efficient.*

Proof in Appendix B.1.

This result arises because the private cost of green entry κ_g dominates any productivity gains, making the non-green allocation efficient.

2.4.2 Planner's Problem With Environmental Externalities

When each green match generates an additional benefit $\phi > 0$ accruing to society, the planner maximizes:

$$\max_{\{\pi, v_g, v_n\}} L(\pi, v_g, v_n) = \underbrace{\pi \frac{v_g}{\pi + v_g}\left[(p - z) + \phi\right]}_{\text{Green matches with externality}} + \underbrace{(1-\pi)\frac{v_n}{(1-\pi) + v_n}(p - z)}_{\text{Non-green matches}}$$
$$- c(v_g + v_n) - \pi \kappa_g. \quad (4)$$

In the decentralized equilibrium, this externality is not internalized by workers or firms, leading to underinvestment in green employment.

[5] All proofs and derivations are provided in Appendix B.

Proposition 3. *With $\phi > 0$, efficiency requires three alignment conditions:*

1. *Hosios condition: $\eta^{g*} = \frac{v_g^*}{\pi^* + v_g^*} = \eta^{n*} = \frac{v_n^*}{(1-\pi^*) + v_n^*} = \eta$, i.e. bargaining power equals the matching elasticity in both sectors.*
2. *Firm-side alignment: $\phi = \dfrac{\tau_g + \tau_n}{1 - \eta}$.*
3. *Worker-side alignment: $\phi = \dfrac{\tau_g + \tau}{\eta(1 - s_g)}$.*

Proof in Appendix B.2.

In addition to the standard Hosios condition, efficiency requires that policy instruments (τ, τ_g, s_g) jointly align private and social returns for both firms and workers. Because these instruments are interdependent, not all combinations decentralize the planner's allocation. Comparative statics in the one-shot model illustrate these policy channels. To my knowledge, this is the first paper to characterize such efficiency conditions in a search-and-matching model with environmental externalities.

I now turn to the dynamic version of the model, which incorporates unemployment dynamics and policy financing over time.

2.5 Dynamic Model: Extended DMP Framework

Building on the static framework, the dynamic model introduces time and sectoral dynamics to analyze green labor market transitions more comprehensively. This extension permits a unified treatment of unemployment dynamics, funding requirements, and aggregate welfare—central issues for policy design in the green transition.

2.5.1 Physical Environment

Time is discrete, $t = 0, 1, 2, \ldots$. Agents discount at rate $\beta \in (0, 1)$. The labor force is normalized to 1. The economy features two segmented labor markets, green (g) and non-green (n). Existing matches are destroyed at rate $\lambda \in (0, 1)$. The government provides a production subsidy τ_g to green firms financed by a tax τ_n on non-green firms. Workers entering the green sector incur a recurring entry cost κ_g, partially offset by a subsidy $s_g \geq 0$; entry into the non-green sector is costless.

2.5.2 Matching Function

Matching in each sector $j \in \{g, n\}$ uses the same constant-returns (CRS) harmonic-mean technology as in the static benchmark:

$$m_j(u_j, v_j) \;=\; \delta_j \, \frac{u_j v_j}{u_j + v_j},$$

where u_j and v_j denote unemployed workers and vacancies, and δ_j is sector-specific matching efficiency. The implied meeting probabilities are

$$\alpha_{wj} \;=\; \frac{m_j}{u_j} \;=\; \delta_j \, \frac{v_j}{u_j + v_j}, \qquad \alpha_{fj} \;=\; \frac{m_j}{v_j} \;=\; \delta_j \, \frac{u_j}{u_j + v_j}, \qquad j \in \{g, n\}. \tag{5}$$

2.5.3 Discussion of Modeling Choices and Empirical Relevance

Before I move on to the full quantitative analysis, I want to discuss some modeling choices.

Definition of green jobs: Green jobs are defined following Curtis and Marinescu (2022) as employment in renewable energy sectors (solar, wind, EVs), aligning with policy focus under the IRA and the fact that energy-related activities account for roughly 70% of U.S. anthropogenic emissions (World Nuclear Association, 2024). This operational definition complements broader notions (e.g., International Labor Organization (2024)) and task-based classifications (e.g., O*NET; Vona, Marin, Consoli, and Popp, 2018; Consoli, Marin, Marzucchi, and Vona, 2018; Vona, Marin, and Consoli, 2019) while capturing emerging occupations (e.g., solar installers).

Worker entry costs. The parameter κ_g represents the effective cost faced by a worker searching or transitioning into the green sector. In the model, it enters as a per-period flow cost borne by unemployed workers who choose to enter the green market. This reduced-form formulation captures a range of structural barriers observed in practice:

- *Training Costs:* Green jobs often demand new technical and managerial skills, requiring reskilling even in related occupations (Vona et al., 2018; Consoli et al., 2018).
- *Relocation Costs:* Green jobs are geographically dispersed, necessitating relocation from fossil fuel hubs to regions rich in renewable resources (Brookings Institution, 2022; Johnson and Schulhofer-Wohl, 2019; Lim et al., 2023).
- *Unionization and Benefits:* Fossil fuel jobs typically offer stronger union representa-

tion and better benefits compared to green jobs, deterring workers from transitioning (Emden and Murphy, 2019; Pollin, Garrett-Peltier, et al., 2020).

- *Uncertainty and Behavioral Barriers:* Perceived instability in green jobs and behavioral factors such as risk aversion or sunk costs further impede transitions (Villas-Boas, 2021; Dixit and Rob, 1994).

While the model remains agnostic about the precise composition of κ_g, it can be interpreted as the expected monthly cost of remaining in the transition state. Figure 3 shows demand outpacing supply on an indexed basis, consistent with a supply-side constraint on worker entry. We therefore model κ_g as the wedge that keeps potential movers from entering, and use s_g to relax that constraint alongside firm-side incentives.

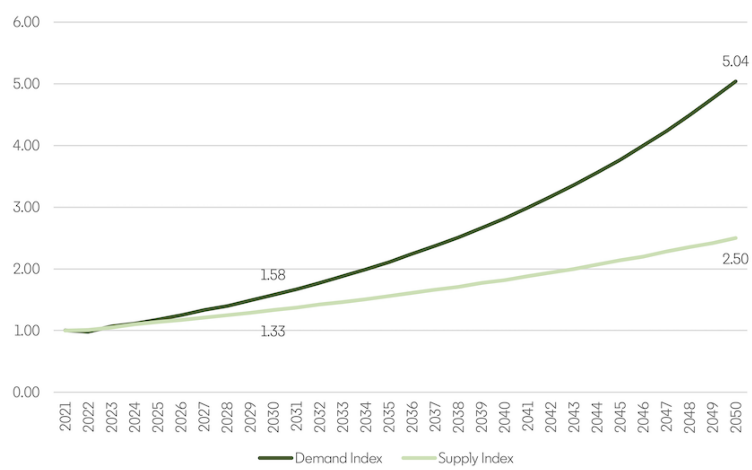


Figure 3. Green talent demand and supply (projected, 2025 onward). *Indexed to 2021; median values and CAGR used for projections.* Dark line: *demand index*; light line: *supply index*. By 2050, demand ≈ 5.0 vs. supply ≈ 2.5, indicating a widening gap. y-axis reports indexed values (2021=1).

Source: LinkedIn Economic Graph (2024).

Firm subsidies, τ_g: The Inflation Reduction Act (IRA) provides substantial production and investment tax credits (τ_g) to incentivize renewable energy deployment, with credits ranging from \$5/MWh to \$32/MWh based on criteria such as labor conditions and domestic content (Bistline et al., 2023). These subsidies have accelerated green technology adoption but have largely overlooked workforce development (Walsh, 2023). By modeling τ_g, the framework captures the financial incentives driving renewable energy expansion while highlighting the gap in policies targeting the labor market.

Why policy-driven solutions? Unlike gradual, largely market-led transitions (e.g., automation or trade), the clean-energy transition is policy-anchored and time-bound. Meeting decarbonization goals requires rapid reallocation that can outpace organic adjustment and historical restructurings show the risks of such unmanaged shifts (Oei et al., 2020). Policy is thus essential to align incentives, overcome coordination frictions, and limit transitional unemployment.

2.5.4 Analysis of the Model

I derive equilibrium conditions linking policy instruments to Beveridge curves, value functions, and wages. This structure permits quantitative evaluation of green employment targets and the associated unemployment and fiscal paths.

Beveridge curves: Let u denote total unemployed workers who choose to enter either the green (u_g) or non-green (u_n) sector, where they may find a job and transition to employment (e_g and e_n, respectively). Similarly, employed workers in both sectors face the possibility of job destruction at rate λ, which returns them to unemployment. The worker-flow diagram in Figure 4 implies steady-state relationships:

$$\alpha_{wg} u_g = \lambda e_g, \qquad \alpha_{wn} u_n = \lambda e_n, \qquad u_g + u_n + e_g + e_n = 1. \tag{6}$$

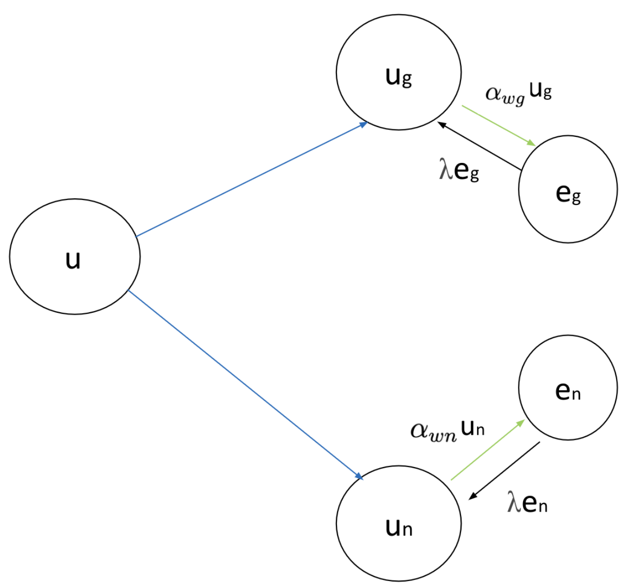


Figure 4. Worker flows between unemployment and employment in both sectors.

Firms' value functions and free entry: Firms choose between entering the green or non-green sectors based on expected profits, which is analogous to free entry in both sectors. Let V represent the value of a vacant firm, and $V_g(J_g)$ and $V_n(J_n)$ be the values of vacant (filled) jobs in the green and non-green sectors, respectively. Each value function is then:

$$V = \max\{V_g, V_n\}, \tag{7}$$

$$V_g = -c \; + \; \beta\big[\alpha_{fg} J_g + (1-\alpha_{fg})V\big], \tag{8}$$

$$V_n = -c \; + \; \beta\big[\alpha_{fn} J_n + (1-\alpha_{fn})V\big], \tag{9}$$

$$J_g = (p+\tau_g) - w_g \; + \; \beta\big[\lambda V + (1-\lambda) J_g\big]^6, \tag{10}$$

$$J_n = p - \tau_n - w_n \; + \; \beta\big[(1-\lambda) J_n + \lambda V\big]. \tag{11}$$

Free entry in both sectors implies $V_g = V_n = V = 0$, yielding

$$c = \beta\alpha_{fg} J_g, \qquad c = \beta\alpha_{fn} J_n. \tag{12}$$

Imposing free entry to the value functions for filled firms also gives:

$$J_g = (p+\tau_g) - w_g + \beta(1-\lambda)J_g, \qquad J_n = p - w_n - \tau_n + \beta(1-\lambda)J_n. \tag{13}$$

Workers' value functions and optimal sector choice: Let U represent the value of an unemployed worker, with $U_g(W_g)$ and $U_n(W_n)$ denoting the values of unemployed (employed) workers in the green and non-green sectors, respectively. Then the value functions are:

$$U = \max\{U_g, U_n\}, \tag{14}$$

$$U_g = z - (\kappa_g - s_g) \; + \; \beta\big[\alpha_{wg} W_g + (1-\alpha_{wg})U_g\big], \tag{15}$$

$$U_n = z \; + \; \beta\big[\alpha_{wn} W_n + (1-\alpha_{wn})U_n\big], \tag{16}$$

$$W_g = w_g \; + \; \beta\big[(1-\lambda)W_g + \lambda U\big], \tag{17}$$

$$W_n = w_n \; + \; \beta\big[(1-\lambda)W_n + \lambda U\big]. \tag{18}$$

[6]Since I normalize productivity p to 1 in the quantitative analysis, the green subsidy τ_g can be interpreted either as an additive transfer to firm revenue or equivalently as a per-unit output subsidy.

Solving yields

$$U_g = \frac{[1-\beta(1-\lambda)]\big(z-(\kappa_g - s_g)\big)+\beta\alpha_{wg}w_g}{(1-\beta)\big(1-\beta(1-\alpha_{wg}-\lambda)\big)}, \quad U_n = \frac{[1-\beta(1-\lambda)]z+\beta\alpha_{wn}w_n}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)}.$$

In equilibrium, workers are indifferent across sectors:

$$U_g = U_n,$$

$$\implies \frac{[1-\beta(1-\lambda)](z-(\kappa_g - s_g))+\beta\alpha_{wg}w_g}{(1-\beta)(1-\beta(1-\alpha_{wg}-\lambda))} = \frac{[1-\beta(1-\lambda)]z+\beta\alpha_{wn}w_n}{(1-\beta)(1-\beta(1-\alpha_{wn}-\lambda))}. \tag{19}$$

which pins down the endogenous composition of search across g and n.

2.5.5 Bargaining Problems

Non-green sector: Let us start by describing the terms of trade in a meeting between a firm and an unemployed in the non-green sector. Solving the standard Nash bargaining problem leads to the following condition that must be satisfied:

$$(1-\eta)(W_n - U_n) = \eta J_n, \tag{20}$$

This condition indicates that each party receives a share of the total surplus from the match, proportional to their bargaining power. (Recall that η represents the worker's bargaining power.) Substituting the value functions J_n, U_n, and W_n from above equations (12), (17), and (19) respectively, I can express the wage in the non-green sector as follows:

$$w_n = \frac{(1-\eta)z\,[1-\beta(1-\lambda)]+\eta(p-\tau_n)\,[1-\beta(1-\lambda-\alpha_{wn})]}{1-\beta(1-\lambda-\eta\alpha_{wn})}. \tag{21}$$

(See Appendix C.1.) The wage depends on tightness through α_{wn} and falls with the tax τ_n.

Green sector: Next, consider the bargaining problem between a firm and a worker in the green sector, where workers incur real costs to enter the sector, while firms receive government subsidies when they match with workers and produce. Again, I have:

$$(1-\eta)\big(W_g - U_g\big) = \eta J_g,$$

which yields

$$w_g = \frac{(1-\eta)\big(z-(\kappa_g-s_g)\big)\,[1-\beta(1-\lambda)] + \eta(1+\tau_g)p\,[1-\beta(1-\lambda-\alpha_{wg})]}{1-\beta(1-\lambda-\eta\alpha_{wg})}. \tag{22}$$

(See Appendix C.2.) The wage rises with firm- and worker-side support (τ_g and s_g) and declines with higher entry costs κ_g.

2.5.6 Government Budget Constraint

Tax revenues from the non-green sector finance subsidies to green firms and workers:

$$\underbrace{\tau_n \cdot \alpha_{fn} v_n}_{\text{Tax revenue from non-green firms}} = \underbrace{\tau_g \cdot \alpha_{fg} v_g}_{\text{Subsidies to green firms}} + \underbrace{s_g \cdot u_g}_{\text{Subsidies to green workers}}.$$

$$\tau_n = \frac{\tau_g\,\alpha_{fg} v_g + s_g\,u_g}{\alpha_{fn} v_n}, \tag{23}$$

where α_{fj} are firm matching rates and u_g, v_g, v_n denote the relevant stocks.

2.5.7 Definition of Steady-State Equilibrium

A steady state equilibrium of the model consists of wages (w_g, w_n) for workers in the green and non-green sectors, a green production subsidy τ_g, a flat tax τ_n paid by non-green firms, measures of vacancies in both sectors (v_g, v_n), and measures of employed and unemployed workers in the various states (u_g, u_n, e_g, e_n). The subsidy and tax satisfy the government budget constraint (23). The remaining equilibrium variables satisfy two free entry conditions (12), two wage curves (21, 22), three Beveridge curves (6), and optimal entry condition (19), after one replaces the various α's with the respective matching probabilities given in (**??**).

3 Calibration

I calibrate the benchmark model to the U.S. economy in 2022, focusing on labor market dynamics in the green transition. One model period corresponds to one calendar month. Parameters with direct empirical counterparts are set exogenously. The discount factor is $\beta = 0.9959$, consistent with a 5% annual interest rate. Worker productivity p is normalized to 1, and the matching function is assumed to exhibit constant returns to scale ($\psi = 0$).

Following Shimer (2005), workers' bargaining power is set to $\eta = 0.72$, and the non-employment benefit to $z = 0.4p$ (40% of average productivity).

Several calibration targets discipline the model. First, I target a 2% wage premium for green workers relative to non-green workers, based on IMF staff estimates (Fund, 2022).[7] Second, the green employment share is set at 2% of the total U.S. workforce, consistent with 2022 estimates from the Energy Information Administration ((EIA), 2024) and calculation of employment share in the renewable sector[8]. Third, the hiring likelihood ratio for workers with green skills relative to others is set to 29%, as reported by LinkedIn's *Green Skills Report* (LinkedIn Economic Graph, 2023). Fourth, overall labor market tightness, measured as the ratio of vacancies to unemployed workers, is set to 1.868, consistent with FRED data (FRED Blog, 2024). Finally, the unemployment rate is targeted at 3.5%, matching BLS data for 2023 (of Labor Statistics (BLS), 2023).

I also incorporate the scale of green subsidies under the Inflation Reduction Act (IRA). Production and investment tax credits range from \$5/MWh to \$32/MWh depending on eligibility (Bushnell and Smith, 2024). With 0.91 billion MWh of renewable generation in 2022 (U.S. Department of Energy, 2023) and GDP of \$25.44 trillion (Bank, 2022), these subsidies correspond to 0.018–0.114% of GDP. I set the benchmark subsidy rate at 0.1% of GDP.

The model matches the targeted moments closely, as shown in Table 1.

Moment	**Model**	**Target**
Wage premium (green/non-green)	1.02	1.02
Employment share (green)	0.0195	0.0195
Hiring likelihood ratio	1.29	1.29
Labor market tightness	1.868	1.868
Unemployment rate	3.5%	3.5%

Table 1: Model performance in matching calibration targets.

The internally calibrated parameters needed to match these moments are reported in Table 2.

[7] Other studies find larger premiums, e.g., about 4% in VoxEU (2023) and up to 20% in Curtis and Marinescu (2022), but I adopt the conservative estimate.

[8] In 2022, there were 3.3 million renewable energy jobs ((E2), 2023) and 212.4 million total jobs ((REA), 2024), resulting in $e_g/e \approx 2\%$.

Parameter	Description	Value
c	Vacancy creation cost	0.1223
λ	Separation rate	0.0192
δ_g	Matching efficiency (green)	0.8476
δ_n	Matching efficiency (non-green)	0.8165
κ_g	Worker entry cost (green)	0.7146

Table 2: Internally calibrated parameters.

Identification: The separation rate λ is calibrated to match the steady-state unemployment rate (3.5%), the vacancy cost c to aggregate tightness $(\theta = 1.87)$, the worker entry cost κ_g to the green employment share (2%), and the relative matching efficiency parameters (δ_g, δ_n) to replicate the hiring-likelihood ratio (1.29) and the green wage premium (1.8%). This calibration anchors the model to key features of the U.S. labor market in 2022 and provides a benchmark for policy experiments. With the model successfully reproducing the wage premium, employment shares, hiring patterns, tightness, and unemployment rate, I am equipped to address the central policy question: *Which mix of worker-side and firm-side interventions can raise green employment from 2% to 14% of the labor force by 2030, as projected by WorkingNation (2024), while minimizing unemployment, costs and welfare distortions?*

4 Quantitative Analysis

The U.S. aims to raise the share of green employment from 2% to 14% of total jobs by 2030, implying nearly 24 million green jobs (WorkingNation, 2024). Using the calibrated model, I evaluate the policy levers needed to achieve this target, focusing on two margins: (i) lowering worker entry costs into the green sector (κ_g, via subsidies s_g) and (ii) raising production subsidies for green firms (τ_g).

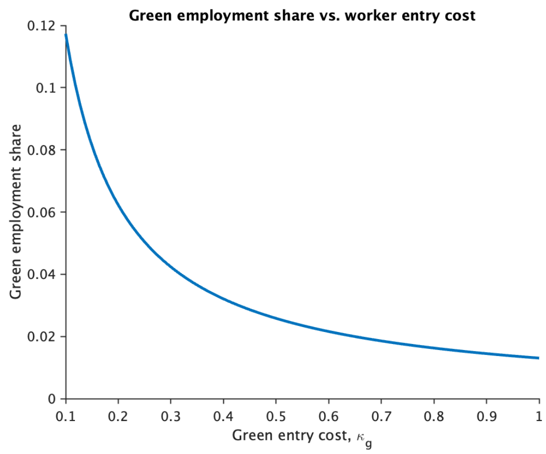


Figure 5. Lower entry costs raise green employment share.

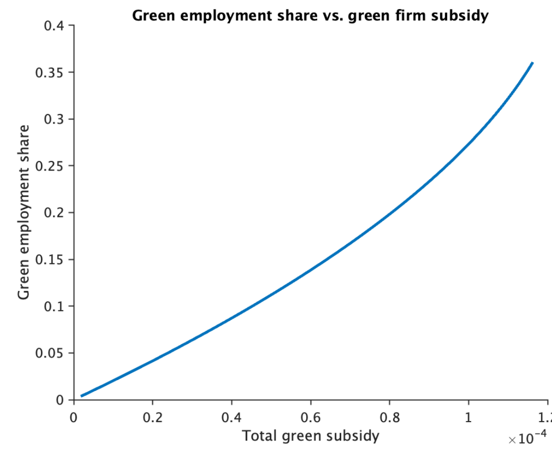


Figure 6. Higher firm subsidies raise green employment share.

4.1 Reaching the Green Employment Target

Figure 5–6 illustrate how worker- and firm-side instruments each expand green employment. Table 3 summarizes alternative policy mixes that attain the 14% green employment target. Although all strategies succeed in principle, they differ markedly in composition and cost.

Table 3: Alternative Policy Mixes Achieving 14% Green Employment

Policy configuration	**Worker subsidy** *(% of GDP)*	**Firm subsidy** *(% of GDP)*	**Total subsidy** *(% of GDP)*	**Green employment** *(%)*
Dual intervention	0.48	0.07	0.55	14.0
Worker-only policy	0.60	0.00	0.60	14.0
Firm-only policy	0.00	0.58	0.58	14.0
Baseline (2022)	0.00	0.00	0.00	2.0

Note: All policy configurations reach an equilibrium green employment share of 14%, starting from a 2% baseline. Subsidy ratios are expressed as percentages of equilibrium GDP.

The baseline economy features only 2% of total employment in the green sector. All three policy paths can reach the 14% target, but they differ in the composition of fiscal support. The combined (dual) strategy achieves the target with the smallest total subsidy burden, indicating complementarity between firm incentives and worker entry support.

4.2 Welfare, Fiscal Cost, and Unemployment Effects

Table 4 compares the resulting equilibria along welfare, fiscal cost, and unemployment dimensions. The dual-subsidy policy dominates: it yields the highest welfare, the lowest unemployment rate, and the smallest fiscal cost relative to GDP.

Table 4: Policy Trade-offs: Unemployment, Fiscal Cost, and Welfare

Policy configuration	Unemployment *(%)*	Fiscal cost *(% of GDP)*	Welfare *(level)*
Dual intervention	**4.77**	**0.55**	**0.962**
Worker-only policy	5.16	0.60	0.959
Firm-only policy	4.80	0.58	0.960
Baseline (2022)	3.50	0.00	0.970

Compared with a worker-only subsidy, the dual policy increases welfare by 0.31%, reduces fiscal cost by 0.05 pp (5 bps) of GDP, and lowers unemployment by 0.39 pp (39 bps). Relative to a firm-only policy, welfare is higher by 0.21%, fiscal cost is lower by 0.03 pp (3 bps), and unemployment is lower by 0.03 pp (3 bps). Welfare under the dual policy nevertheless remains below the baseline and unemployment higher, reflecting the reallocation–productivity trade-off in the model. In the baseline DMP framework, green jobs generate no external benefits; thus, shifting labor toward the higher-wage green sector lowers average productivity even as employment goals are met. Following the climate-macro literature, the next section introduces a positive externality from green employment into the welfare function to assess how such spillovers affect aggregate efficiency and the net welfare effects of targeted subsidies.

4.3 Welfare with Green Externalities

I incorporate a positive externality from green employment into the welfare function. The standard DMP welfare is

$$W_0 \;=\; p(e_g + e_n) \;+\; (z - \kappa_g)u_g \;+\; zu_n \;-\; c(v_n + v_g),$$

and adding a per-job externality ϕ (the environmental benefit per green job) gives

$$W_\phi \;=\; p(e_g + e_n) \;+\; (z - \kappa_g)u_g \;+\; zu_n \;-\; c(v_n + v_g) \;+\; \phi\, e_g.$$

4.3.1 Threshold Analysis

The welfare gain or loss from policy interventions depends critically on the value of ϕ. Figure 7 shows how welfare depends on ϕ. Under the dual policy, welfare equals the baseline ($W_\phi = W_0$) when $\phi^\star$. Thus, only modest externalities (bigger than $\phi^\star = 0.062$) are required for green reallocation to become welfare-improving.

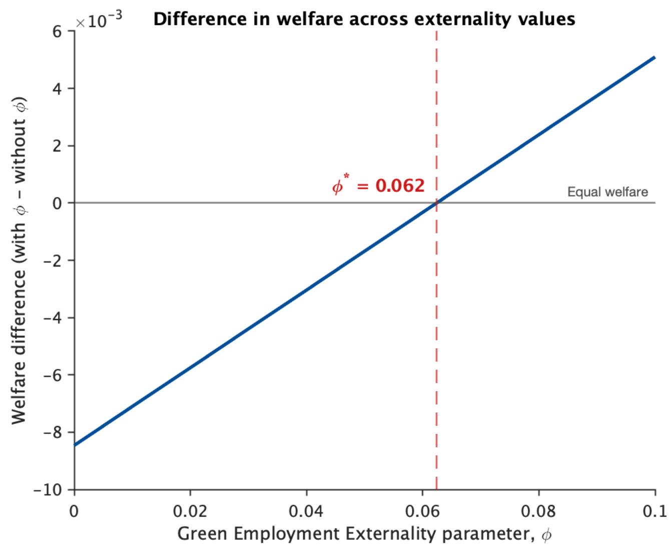


Figure 7. Welfare gains as a function of the externality parameter ϕ.

4.3.2 Output-Equivalent Interpretation

To benchmark the magnitude of $\phi^\star$, I translate it into an *output-equivalent* change i.e., the percent increase in model output that delivers the same welfare gain. With output defined as Aggregate Output $= p(e_g + e_n)$, the implied output increase at $\phi^\star$ is 0.8752%. Thus, the externality needed to meet the welfare target is modest: expanding the green sector is welfare-improving once its environmental benefits are worth at least a 0.88% increase in output. This offers a simple, transparent benchmark for quantifying externalities in DMP-style labor market models.

Here's a cleaner, more "human" (but still academic) version you can paste in. I've removed the awkward parentheses, tightened phrasing, and kept notation consistent with LaTeX.

4.4 Transition to a 14% Green Share

I implement the transition as a sequence of policy–induced steady states that hit intermediate adoption targets. For each target green employment share $L_g \in \{2\%, 4\%, \ldots, 14\%\}$, I solve for the welfare–maximizing mix of worker-entry support (s_g) and firm-side production subsidies (τ_g) subject to equilibrium conditions and a balanced budget. The next target uses the previous solution as a warm start, generating a controlled path of (s_g, τ_g), allocations, and prices indexed by the achieved L_g. For each L_g, I record unemployment, total funding as a share of GDP, and its worker/firm decomposition, and I track green tightness, the job-finding rate, and the vacancy-filling rate along the path.[9]

This transition study design mirrors how climate policy is actually made. Under the Paris Agreement, countries submit and update Nationally Determined Contributions on a five-year cycle, with each submission required to represent a progression in ambition; major roadmaps similarly operationalize long-run goals via dated waypoints, e.g., the IEA's Net Zero by 2050 sets out 400+ milestones (many by 2030/2035) rather than a single 2050 endpoint.[10] In this institutional environment, the operative objects for policymakers are the least-cost instrument mix and fiscal burden at each interim target, not a single perfect-foresight off–steady-state path. The sequence-of-steady-states approach therefore yields the policy mix and funding profile at the milestones that governments actually set and monitor.

Figure 8 plots unemployment and total funding (percent of GDP) against achieved L_g. The response is modest and hump-shaped: unemployment rises from about 4.55% at $L_g \approx 4\%$ to a peak near 4.75% around $L_g = 8$–10%, then eases as the target is approached. Funding follows a similar arc, increasing from roughly 0.53% of GDP at low L_g to a peak near 0.58% mid-transition before tapering off. Figure 9 shows the fiscal mix rotating with scale: worker support accounts for a larger share early and late in the path, while firm subsidies do more of the work midway at $L_g \approx 6$–10%.

[9] With matching $m_g = \delta_g\, u_g v_g/(u_g + v_g)$ and green tightness $\theta_g \equiv v_g/u_g$, the rates are $f(\theta_g) \equiv m_g/u_g = \delta_g\, \theta_g/(1+\theta_g)$ and $q(\theta_g) \equiv m_g/v_g = \delta_g/(1+\theta_g)$. Hence $f'(\theta_g) = \delta_g/(1+\theta_g)^2 > 0$ and $q'(\theta_g) = -\delta_g/(1+\theta_g)^2 < 0$. In the figures, "pathwise elasticities" are computed as finite differences of $f(\theta_g)$ and $q(\theta_g)$ with respect to θ_g along the sequence of steady states and serve as diagnostics for which margin (entry vs. vacancies) is locally more responsive.

[10] See UNFCCC, The Paris Agreement (five-year cycle; successive NDCs represent a "progression") and NDC explainer pages; International Energy Agency, Net Zero by 2050 (and 2023 update) compiling 400+ sectoral and technology milestones.

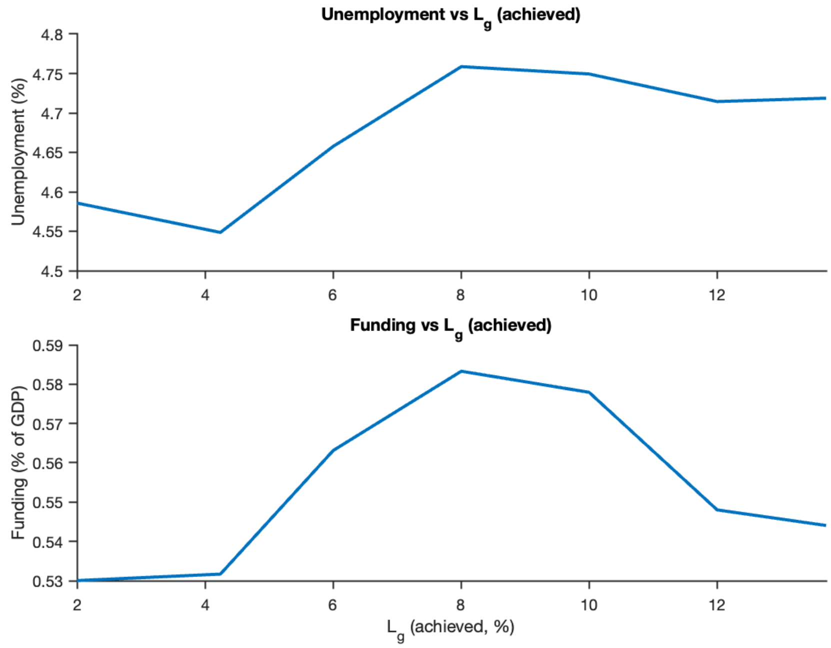


Figure 8. Unemployment and total funding vs. achieved L_g on the welfare-maximizing path to 14%.

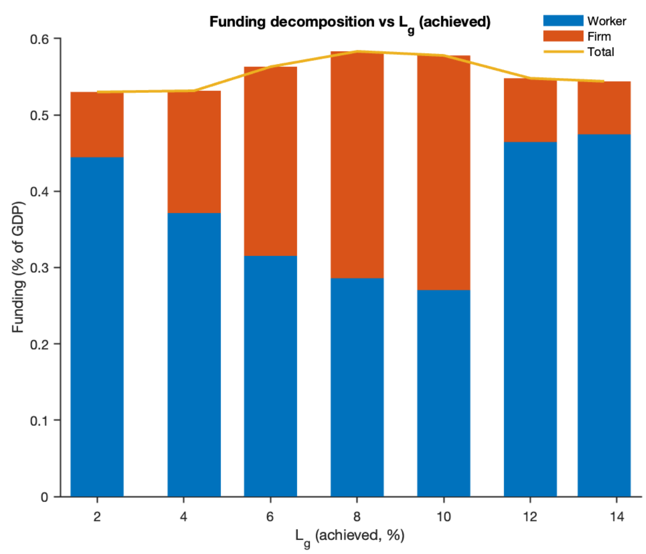
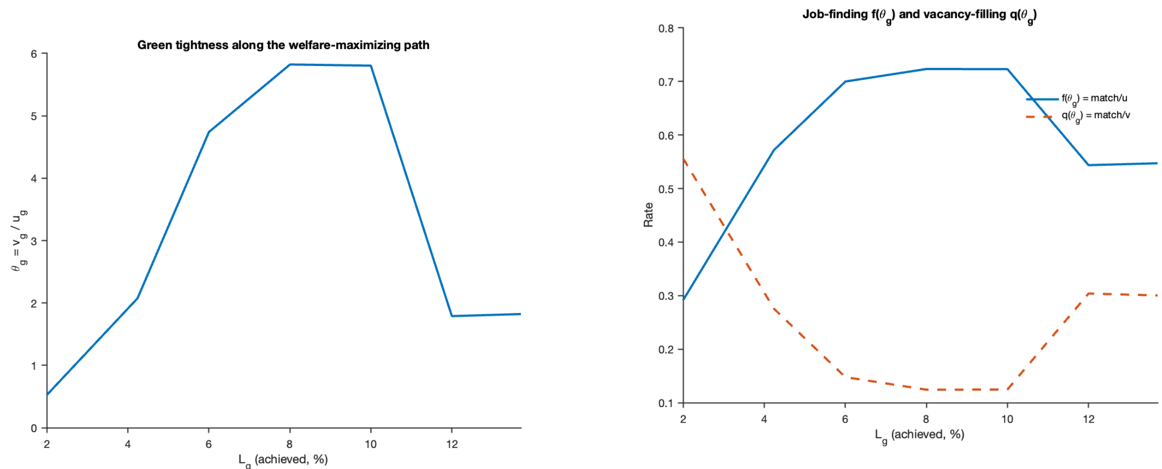


Figure 9. Funding decomposition (worker vs. firm) along the same path; line shows total funding as a percent of GDP.

Mechanism: The hump and the rotation are driven by the evolution of green tightness θ_g and the implied local elasticities of the two matching margins. Early on, vacancies outpace entrants, θ_g jumps, $f(\theta_g)$ is steep and $q(\theta_g)$ remains high, so extra support to workers who are the one paying the entry cost generates matches at low unemployment and low marginal cost. Midway, θ_g becomes very high: $f(\theta_g)$ flattens mechanically and

$q(\theta_g)$ is small, making both margins locally inelastic; additional matches are most expensive here, producing the hump in unemployment and funding. Late in the path, policy tilts back toward worker entry, θ_g falls, vacancy filling improves (higher $q(\theta_g)$), and both unemployment and funding ease. Figures 10–11 document these dynamics.



(a) Green tightness $\theta_g \equiv v_g/u_g$.

(b) Job-finding $f(\theta_g) = \text{match}_g/u_g$ and vacancy-filling $q(\theta_g) = \text{match}_g/v_g$.

Figure 10. Tightness and matching rates along the welfare-maximizing path.

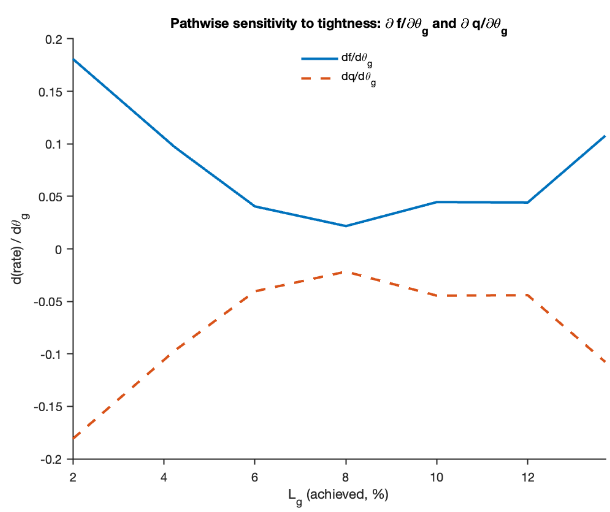


Figure 11. Pathwise sensitivities $\partial f/\partial\theta_g$ and $\partial q/\partial\theta_g$ (finite differences).

Transition policy insights: Design and timing both matter. Dual-sided support is more efficient than one-sided alternatives at fixed L_g, but the emphasis should be state-contingent: front- and back-load worker support when entry is binding (low and high L_g), and lean more on firm subsidies in the tight, match-inelastic middle. The mid-path rise in unemployment and funding is not a design flaw; it reflects the matching block when both job-finding and vacancy-filling become locally unresponsive. Recognizing this structure clarifies budgeting, helps communicate that mid-path cost pressures are temporary, and guides the sequencing of instruments to keep marginal matching costs contained.

5 Economic Insights

This section distills the mechanisms that govern efficient green labor reallocation and the policy principles they imply. The results are presented as qualitative design rules that arise from two-sided search with a entry cost on workers, production subsidies to firms, and a balanced-budget instrument set. The calibration illustrates magnitudes; the logic is general.

In this setup, reallocation is governed by a real-world-like misalignment of private costs and benefits: workers pay an entry cost to become green entrants, while firms receive a production subsidy that is only realized if a vacancy meets a worker and produces. Workers' entry thus creates benefits that accrue partly to firms (by raising the value of a filled vacancy), and firms' vacancy creation generates benefits that accrue partly to workers (by raising the return to entry), neither of which is fully internalized. Left to itself, each margin under-responds and the transition stalls. A dual policy that simultaneously lowers worker entry costs and supports vacancy creation realigns private and social returns on both sides, restores complementarity, and increases matches per unit of public spending under a balanced budget.

Insight 1: Dual beats single on the margins that matter. Worker-only or firm-only policies can push the target share up, but they do so by leaving the other side as the bottleneck. The consequence is avoidable unemployment (idle search meets empty vacancies or vice versa) and a higher funding burden (each dollar of subsidy buys few additional matches). By contrast, a coordinated dual policy aligns entry and vacancy creation, increasing match formation at lower fiscal cost. In our benchmark, the same 14 percent target is reached with materially lower unemployment and lower cumulative outlays than the best one-sided alternative. This static dominance is consistent with two-sided market logic: when outcomes depend on complementary participation, subsidizing both sides outperforms subsidizing one.

Insight 2: Dynamic design rule: hump-shaped intensity and rotating composition. Optimal policy is state contingent. Along the adoption path from 2 to 14 percent, the total support required is hump-shaped: modest early on, rising at intermediate adoption, and easing near the target. The composition also rotates over the path: worker support is most effective early and late, while firm-side support is most effective in the middle. The economic forces are:

- **Early stage:** The dominant barrier is the workers' entry cost. The pool of potential movers is still large, and firms can absorb additional entrants. A modest nudge to worker entry generates many new matches per dollar, while firms support plays a secondary role.

- **Middle stage:** As reallocation proceeds, the green market becomes temporarily tight. Both margins turn locally less responsive, i.e. workers require stronger inducements and firms face congestion, so total support peaks. With the binding constraint on filling vacancies in a tight market, firm-side subsidies are especially effective.

- **Late stage:** The sector has scaled, so additional vacancies match more easily, but the remaining pool of potential movers is thin. The binding constraint swings back to new entry, and worker support again delivers the highest return. The optimal mix thus follows a worker–firm–worker sequence over the transition.

Insight 3: When does the transition raise welfare? Welfare hinges on a simple productivity–externality tradeoff: shifting a worker into green activity entails a marginal private output loss (training/switching costs, mismatch) but generates a non-internalized environmental gain. Mapping that gain into output units yields a clear rule: the transition is welfare-improving whenever the external benefit per additional green match exceeds the productivity-equivalent cost of creating that match. In the benchmark, an output-equivalent benefit of about 0.88 percent of output suffices; the exact value is model-specific, but the principle that external benefits must exceed marginal reallocation costs does not depend on calibration.

6 Conclusion

This paper studies how to design policy for large, two-sided reallocations and applies the framework to the green transition. In a two-sector DMP model with costly worker entry and firm-side hiring incentives that stimulate vacancy creation, I compare worker-only, firm-only, and coordinated dual policies on three margins central to success: unemployment, fiscal cost, and welfare.

Three results emerge. First, dual beats single on the margins that matter: relative to the best one-sided alternative, a coordinated policy that supports both worker entry and vacancy creation attains the same 14% target with 39 bps lower unemployment and 5 bps of GDP lower fiscal cost. Second, optimal policy is dynamic: total support is hump-shaped over the adoption path, and its composition rotates: worker support is most effective early and late, while firm-side support is most effective mid-transition. Third, when environmental benefits are modeled as an output-equivalent externality, a threshold of 88 bps of GDP is sufficient for the coordinated policy to raise aggregate welfare relative to no intervention.

The policy design rule is simple and portable: use a dual policy, scale its intensity over the path, and rotate its composition as the binding margin shifts; assess welfare by comparing output-equivalent external benefits to the measured fiscal cost of support. Since the mechanism is two-sided search with costly entry and vacancy creation, the same rule applies to other large-scale reallocations—digital upskilling and AI adoption, supply-chain rebuilding—where getting the design right, rather than merely intervening, determines whether the shift is slow and expensive or fast and financially sustainable.

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A One-shot DMP Model

This section provides the detailed derivations for the static (one-period) benchmark version of the model, which is useful to illustrate the role of coordination frictions and the operation of policy instruments in segmented labor markets.

A.1 Bargaining

Green Sector: In the green sector, wages are determined via Nash bargaining between firms and workers. The problem is

$$\begin{aligned}
&\max_{w_g}(w_g - z)^\eta \,(p + \tau_g - w_g)^{1-\eta}, \\
&\implies \frac{\eta}{w_g - z} = \frac{1-\eta}{p + \tau_g - w_g}, \\
&\therefore\; w_g = \eta(p + \tau_g) + (1-\eta)z.
\end{aligned}$$

The bargained wage in the green sector is increasing in productivity p and the production subsidy τ_g, while the fallback option z provides the lower bound.

Non-Green Sector: In the non-green sector, the structure of the bargaining problem is analogous, except that firms pay a per-unit tax τ_n:

$$\begin{aligned}
&\max_{w_n}(w_n - z)^\eta \,(p - \tau_n - w_n)^{1-\eta}, \\
&\implies \frac{\eta}{w_n - z} = \frac{1-\eta}{p - \tau_n - w_n}, \\
&\therefore\; w_n = \eta(p - \tau_n) + (1-\eta)z.
\end{aligned}$$

Relative to the green sector, non-green wages are depressed by the tax burden τ_n.

A.2 Interior Equilibrium with Both Sectors Operating

When both sectors are active, the equilibrium variables are $\{\pi, v_g, v_n, w_g, w_n\}$, where π is the share of (initially unemployed) workers who *enter the green market*. The conditions are:

$$
\begin{aligned}
c &= \frac{\pi}{\pi + v_g}\big(p + \tau_g - w_g\big), \\
c &= \frac{1-\pi}{1-\pi + v_n}\big(p - \tau_n - w_n\big), \\
w_g &= \eta(p + \tau_g) + (1-\eta)z, \\
w_n &= \eta(p - \tau_n) + (1-\eta)z, \\
\tau_n \, \frac{(1-\pi)}{(1-\pi) + v_n} v_n &= \tau_g \, \frac{\pi}{\pi + v_g} v_g \ + \ s_g \, \pi, \\
\pi &= \begin{cases} 0 & \text{if } G < N, \\ \in (0,1) & \text{if } G = N, \\ 1 & \text{if } G > N, \end{cases}
\end{aligned}
$$

where the expected utilities of workers (before matching) are

$$
G \ = \ -(\kappa_g - s_g) + \frac{v_g}{\pi + v_g} \, w_g + \Big(1 - \frac{v_g}{\pi + v_g}\Big) z, \qquad N \ = \ \frac{v_n}{1-\pi+v_n} \, w_n + \Big(1 - \frac{v_n}{1-\pi+v_n}\Big) z.
$$

Notes: (i) The government budget uses *matches* as the base for firm-side flows and *entrants* as the base for worker-side transfers. Since the unit mass of workers is initially unemployed and entry occurs *before* matching, the mass receiving the level subsidy is π. (ii) Throughout, s_g is a *level* (per-period) subsidy; the effective worker cost is $\kappa_g - s_g$.

A.3 Corner Equilibria

The economy can also settle in corner solutions.

A.3.1 Case $\pi = 0$: Non-Green Only

It must be the case that $v_g = 0$ iff $\pi = 0$. In this case, only the non-green labor market operates with $v_n > 0$ and the matching probabilities simplify to

$$
\alpha_{wg} = \alpha_{fg} = 0, \quad \alpha_{wn} = \frac{v_n}{1+v_n}, \quad \alpha_{fn} = \frac{1}{1+v_n}.
$$

Equilibrium conditions are

$$
\begin{aligned}
&v_g = 0, \\
&c = \frac{1}{1+v_n}\big(p - \tau_n - w_n\big), \\
&w_g = \eta(p + \tau_g) + (1-\eta)z, \\
&w_n = \eta(p - \tau_n) + (1-\eta)z, \\
&\tau_n = \tau_g = 0, \\
&\pi = 0 \text{ with } G < N,
\end{aligned}
$$

where G and N denote the (counterfactual) expected utilities:

$$G = -(\kappa_g - s_g) + z, \qquad N = \frac{v_n}{1+v_n} w_n + \Big(1 - \frac{v_n}{1+v_n}\Big) z,$$

so

$$G - N = -(\kappa_g - s_g) - \frac{v_n}{1+v_n}(w_n - z) < 0.$$

A.3.2 Case $\pi = 1$: Green Only

It must be the case that $v_n = 0$ iff $\pi = 1$. In this case, only the green labor market operates with $v_g > 0$ and the matching probabilities are:

$$\alpha_{wn} = \alpha_{fn} = 0, \quad \alpha_{wg} = \frac{v_g}{1+v_g}, \quad \alpha_{fg} = \frac{1}{1+v_g}.$$

Equilibrium conditions are:

$$
\begin{aligned}
&v_n = 0, \\
&c = \frac{1}{1+v_g}\big(p + \tau_g - w_g\big), \\
&w_g = \eta(p + \tau_g) + (1-\eta)z, \\
&w_n = \eta(p - \tau_n) + (1-\eta)z, \\
&\tau_n = \tau_g = 0, \\
&\pi = 1 \text{ with } G > N,
\end{aligned}
$$

with

$$G = -(\kappa_g - s_g) + \frac{v_g}{1+v_g} w_g + \Big(1 - \frac{v_g}{1+v_g}\Big) z, \qquad N = z,$$

so

$$G - N = -(\kappa_g - s_g) + \frac{v_g}{1+v_g}(w_g - z) > 0.$$

B Social Planner's Problem and Efficiency

B.1 Planner without Externality

Consider first the benchmark case without environmental externalities ($\phi = 0$), where the only difference between sectors is that green-sector workers must pay an entry cost κ_g. The planner maximizes

$$L(\pi, v_g, v_n) = \pi \frac{v_g}{\pi + v_g}(p - z) + (1-\pi)\frac{v_n}{(1-\pi) + v_n}(p - z) - c(v_g + v_n) - \pi\kappa_g. \tag{A.1}$$

The FOCs yield

$$\frac{\pi}{\pi + v_g} = \sqrt{\frac{c}{p-z}}, \qquad \frac{1-\pi}{1-\pi+v_n} = \sqrt{\frac{c}{p-z}},$$

so match probabilities are equal across sectors. The worker-allocation condition is

$$\Big(\frac{v_g}{\pi + v_g}\Big)^2(p-z) - \Big(\frac{v_n}{1-\pi+v_n}\Big)^2(p-z) = \kappa_g.$$

This cannot hold with equal probabilities unless $\kappa_g = 0$, so the planner chooses a corner π.

B.2 Planner with Positive Externality ($\phi > 0$)

Now green matches generate an external benefit ϕ. With $\phi > 0$, the planner maximizes:

$$\max_{\{\pi, v_g, v_n\}} L(\pi, v_g, v_n) = \underbrace{\pi \frac{v_g}{\pi + v_g}\big[(p-z) + \phi\big]}_{\text{Green matches with externality}} + \underbrace{(1-\pi)\frac{v_n}{(1-\pi)+v_n}(p-z)}_{\text{Non-green matches}} - c\,(v_g + v_n) - \pi\,\kappa_g.$$

The FOCs imply efficient match probabilities

$$\frac{v_g}{\pi + v_g} = 1 - \sqrt{\frac{c}{(p-z)+\phi}}, \qquad \frac{v_n}{1-\pi+v_n} = 1 - \sqrt{\frac{c}{p-z}}.$$

The worker-allocation condition becomes

$$\Big(\frac{v_g}{\pi+v_g}\Big)^2[(p-z)+\phi]-\Big(\frac{v_n}{1-\pi+v_n}\Big)^2(p-z)=\kappa_g,$$

which yields an interior solution provided ϕ is large enough relative to κ_g.

B.3 Competitive Market with Policy

In equilibrium, workers are indifferent if

$$-(\kappa_g-s_g)+\frac{v_g}{\pi+v_g}(w_g-z)\;=\;\frac{v_n}{1-\pi+v_n}(w_n-z).$$

Wages are Nash outcomes:

$$w_g=\eta(p+\tau_g)+(1-\eta)z,\qquad w_n=\eta(p-\tau_n)+(1-\eta)z.$$

Firms enter until

$$c=\frac{\pi}{\pi+v_g}\big(p+\tau_g-w_g\big),\qquad c=\frac{1-\pi}{1-\pi+v_n}\big(p-\tau_n-w_n\big).$$

Implied green employment is

$$E_g=\pi\frac{v_g}{\pi+v_g}\;=\;\pi\Big(1-\frac{c}{(1-\eta)(p+\tau_g-z)}\Big),$$

where the last equality uses the Nash wage to write the firm surplus in terms of primitives.[11]

Remark (Worker entry condition). Under the level subsidy, entry is governed by the indifference condition with κ_g-s_g (rather than a fractional $(1-s_g)\kappa_g$). Closed-form "alignment" equalities that rely on a multiplicative form do not apply; what matters is the *level* effective cost κ_g-s_g on the worker margin.

B.4 Efficiency Conditions

To decentralize the planner's allocation $\{\pi^*,v_g^*,v_n^*\}$, market conditions must be consistent with the planner's FOCs.

[11] This expression is standard in static DMP with Nash bargaining under CRS and is provided here only to illustrate how τ_g shifts green matches.

1. Firm margin. With Nash bargaining and CRS, the usual Hosios condition equates worker bargaining power to the elasticity of the matching function with respect to vacancies. In the one-shot CRS setting, this can be stated as

$$\eta^{g*} = \frac{v_g^*}{\pi^* + v_g^*}, \qquad \eta^{n*} = \frac{v_n^*}{1 - \pi^* + v_n^*}.$$

Given Hosios, the firm-side externality can be offset by suitable (τ_g, τ_n). A convenient summary is that a higher τ_g (and/or lower τ_n) raises the private firm surplus toward the social marginal benefit (which includes ϕ).

2. Worker margin. On the worker side, decentralization requires that the private entry condition (with effective cost $\kappa_g - s_g$) matches the planner's allocation condition. Under the level subsidy, alignment is achieved by choosing s_g so that the private benefit differential across sectors equals $\kappa_g - s_g$ (rather than a fraction of κ_g). Formally, one equates the market indifference condition above to the planner's worker condition, with ϕ entering only on the green side in the planner's problem.

Summary. Efficiency requires (i) the Hosios condition on both sectors, and (ii) policies (τ_g, τ_n, s_g) that align private surpluses with social surpluses. Under the level subsidy, this alignment uses the *level* effective cost $\kappa_g - s_g$ on the worker margin (rather than a multiplicative form).

B.5 Government Budget

In the one-shot setting, total taxes collected from non-green matches must cover the cost of green subsidies:

$$\tau_n \cdot \underbrace{\frac{(1-\pi)}{(1-\pi) + v_n} v_n}_{\text{non-green matches}} = \tau_g \cdot \underbrace{\frac{\pi}{\pi + v_g} v_g}_{\text{green matches}} + \underbrace{s_g\,\pi}_{\text{entry transfers to workers}}.$$

The worker-side transfer is paid to the mass of *entrants* π (entry occurs prior to matching), whereas the firm-side flows are paid on *matches*.

C Bargaining in the Dynamic Model

This section derives the dynamic Nash bargaining solutions for wages in the non-green and green sectors. I begin from the value functions defined in the main text and proceed step-by-step to obtain closed-form expressions for the wage schedules. Throughout, I maintain the notation introduced earlier: worker bargaining power is $\eta \in (0,1)$, productivity is p, green firm subsidy is τ_g, non-green tax is τ_n, worker green-entry subsidy is s_g, worker green-entry cost is κ_g, discount factor is $\beta \in (0,1)$, and the exogenous separation rate is $\lambda \in (0,1)$. Matching probabilities are α_{wj} for workers and α_{fj} for firms in sector $j \in \{g, n\}$.

C.1 Non-Green Jobs

Proof. The non-green sector value functions (see equations (16), (18), and (11) in the main text) are:

$$U_n = z + \beta\big[\alpha_{wn} W_n + (1-\alpha_{wn})U_n\big], \tag{A.2}$$

$$W_n = w_n + \beta\big[(1-\lambda)W_n + \lambda U_n\big], \tag{A.3}$$

$$J_n = (p - \tau_n - w_n) + \beta\big[(1-\lambda)J_n + \lambda V\big], \tag{A.4}$$

where V is the value of a vacancy and equals zero in equilibrium by free entry.

Step 1: Solving for U_n. Starting from (A.2),

$$U_n = z + \beta\alpha_{wn} W_n + \beta(1-\alpha_{wn})U_n, \tag{A.5}$$

$$U_n - \beta(1-\alpha_{wn})U_n = z + \beta\alpha_{wn} W_n, \tag{A.6}$$

$$\big[1 - \beta(1-\alpha_{wn})\big]U_n = z + \beta\alpha_{wn} W_n, \tag{A.7}$$

so

$$U_n = \frac{z}{1-\beta(1-\alpha_{wn})} + \frac{\beta\alpha_{wn}}{1-\beta(1-\alpha_{wn})} W_n. \tag{A.8}$$

Step 2: Solving for W_n. From (A.3),

$$W_n = w_n + \beta(1-\lambda)W_n + \beta\lambda U_n, \tag{A.9}$$

$$W_n - \beta(1-\lambda)W_n = w_n + \beta\lambda U_n, \tag{A.10}$$

$$\big[1 - \beta(1-\lambda)\big]W_n = w_n + \beta\lambda U_n. \tag{A.11}$$

Substitute (A.8) into (A.11):

$$\big[1-\beta(1-\lambda)\big]W_n = w_n + \beta\lambda\left(\frac{z}{1-\beta(1-\alpha_{wn})} + \frac{\beta\alpha_{wn}}{1-\beta(1-\alpha_{wn})}W_n\right), \tag{A.12}$$

$$\big[1-\beta(1-\lambda)\big]W_n = w_n + \frac{\beta\lambda}{1-\beta(1-\alpha_{wn})}z + \frac{\beta^2\lambda\alpha_{wn}}{1-\beta(1-\alpha_{wn})}W_n. \tag{A.13}$$

Collect terms in W_n on the left:

$$\left(1-\beta(1-\lambda) - \frac{\beta^2\lambda\alpha_{wn}}{1-\beta(1-\alpha_{wn})}\right)W_n = w_n + \frac{\beta\lambda}{1-\beta(1-\alpha_{wn})}z. \tag{A.14}$$

Put over a common denominator $1-\beta(1-\alpha_{wn})$:

$$\frac{\big[1-\beta(1-\lambda)\big]\big[1-\beta(1-\alpha_{wn})\big] - \beta^2\lambda\alpha_{wn}}{1-\beta(1-\alpha_{wn})}\,W_n = w_n + \frac{\beta\lambda}{1-\beta(1-\alpha_{wn})}z. \tag{A.15}$$

Note that

$$\big[1-\beta(1-\lambda)\big]\big[1-\beta(1-\alpha_{wn})\big] - \beta^2\lambda\alpha_{wn} = (1-\beta)\big[1-\beta(1-\alpha_{wn}-\lambda)\big].$$

Therefore,

$$W_n = \frac{w_n\big(1-\beta(1-\alpha_{wn})\big) + \beta\lambda z}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)}. \tag{A.16}$$

Step 3: Substituting back into U_n. Use (A.8) and (A.16):

$$U_n = \frac{z}{1-\beta(1-\alpha_{wn})} + \frac{\beta\alpha_{wn}}{1-\beta(1-\alpha_{wn})}\cdot\frac{w_n\big(1-\beta(1-\alpha_{wn})\big) + \beta\lambda z}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)} \tag{A.17}$$

$$= \frac{z}{1-\beta(1-\alpha_{wn})} + \frac{\beta\alpha_{wn}}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)}\left[w_n + \frac{\beta\lambda}{1-\beta(1-\alpha_{wn})}z\right]. \tag{A.18}$$

Bring terms over a common denominator $(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)$:

$$U_n = \frac{z\,(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)\big(1-\beta(1-\alpha_{wn})\big)} + \frac{\beta\alpha_{wn}w_n}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)} \tag{A.19}$$

$$+\frac{\beta^2\lambda\alpha_{wn}z}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)\big(1-\beta(1-\alpha_{wn})\big)}. \tag{A.20}$$

Using algebra identical to Step 2, this simplifies to

$$U_n = \frac{(1-\beta(1-\lambda))\,z + \beta\alpha_{wn}\,w_n}{(1-\beta)\big(1-\beta(1-\alpha_{wn}-\lambda)\big)}. \tag{A.21}$$

Step 4: Worker surplus. Subtract (A.21) from (A.16):

$$W_n - U_n = \frac{w_n - z}{1-\beta(1-\alpha_{wn}-\lambda)}. \tag{A.22}$$

Step 5: Firm surplus and Nash bargaining. With free entry, $V = 0$; from (A.4),

$$J_n = (p-\tau_n-w_n) + \beta(1-\lambda)J_n, \tag{A.23}$$

$$J_n\big[1-\beta(1-\lambda)\big] = p-\tau_n-w_n, \tag{A.24}$$

$$J_n = \frac{p-\tau_n-w_n}{1-\beta(1-\lambda)}. \tag{A.25}$$

The Nash condition is

$$(1-\eta)(W_n-U_n) = \eta J_n. \tag{A.26}$$

Substitute the surpluses:

$$(1-\eta)\,\frac{w_n-z}{1-\beta(1-\alpha_{wn}-\lambda)} = \eta\,\frac{p-\tau_n-w_n}{1-\beta(1-\lambda)}. \tag{A.27}$$

Solve for w_n by collecting terms:

$$w_n\left[\frac{1-\eta}{1-\beta(1-\alpha_{wn}-\lambda)} + \frac{\eta}{1-\beta(1-\lambda)}\right] = \frac{(1-\eta)\,z}{1-\beta(1-\alpha_{wn}-\lambda)} + \frac{\eta\,(p-\tau_n)}{1-\beta(1-\lambda)}. \tag{A.28}$$

After collecting over the common denominator $1-\beta(1-\lambda-\eta\alpha_{wn})$, the solution is

$$\boxed{w_n = \frac{(1-\eta)\,z\,[1-\beta(1-\lambda)] \;+\; \eta\,(p-\tau_n)\,[1-\beta(1-\lambda-\alpha_{wn})]}{1-\beta\big(1-\lambda-\eta\alpha_{wn}\big)}.} \tag{A.29}$$

Expression (A.29) implies $\partial w_n/\partial\tau_n < 0$, holding tightness (hence α_{wn}) fixed. □

C.2 Green Jobs

Proof. The green sector value functions (see equations (15), (17), and (10) in the main text) are:

$$U_g = z - (\kappa_g - s_g) + \beta\big[\alpha_{wg} W_g + (1-\alpha_{wg})U_g\big], \tag{A.30}$$

$$W_g = w_g + \beta\big[(1-\lambda)W_g + \lambda U_g\big], \tag{A.31}$$

$$J_g = (p + \tau_g - w_g) + \beta\big[(1-\lambda)J_g + \lambda V\big], \tag{A.32}$$

with $V = 0$ by free entry.

Step 1: Solving for U_g. From (A.30),

$$U_g = z - (\kappa_g - s_g) + \beta\alpha_{wg} W_g + \beta(1-\alpha_{wg})U_g, \tag{A.33}$$

$$U_g - \beta(1-\alpha_{wg})U_g = z - (\kappa_g - s_g) + \beta\alpha_{wg} W_g, \tag{A.34}$$

$$\big[1 - \beta(1-\alpha_{wg})\big]U_g = z - (\kappa_g - s_g) + \beta\alpha_{wg} W_g, \tag{A.35}$$

hence

$$U_g = \frac{z - (\kappa_g - s_g)}{1 - \beta(1-\alpha_{wg})} + \frac{\beta\alpha_{wg}}{1 - \beta(1-\alpha_{wg})} W_g. \tag{A.36}$$

Step 2: Solving for W_g. From (A.31),

$$W_g = w_g + \beta(1-\lambda)W_g + \beta\lambda U_g, \tag{A.37}$$

$$W_g - \beta(1-\lambda)W_g = w_g + \beta\lambda U_g, \tag{A.38}$$

$$\big[1 - \beta(1-\lambda)\big]W_g = w_g + \beta\lambda U_g. \tag{A.39}$$

Substitute (A.36) into (A.39):

$$\big[1 - \beta(1-\lambda)\big]W_g = w_g + \beta\lambda\left(\frac{z - (\kappa_g - s_g)}{1 - \beta(1-\alpha_{wg})} + \frac{\beta\alpha_{wg}}{1 - \beta(1-\alpha_{wg})} W_g\right), \tag{A.40}$$

$$\big[1 - \beta(1-\lambda)\big]W_g = w_g + \frac{\beta\lambda}{1 - \beta(1-\alpha_{wg})}\big(z - (\kappa_g - s_g)\big) + \frac{\beta^2\lambda\alpha_{wg}}{1 - \beta(1-\alpha_{wg})} W_g. \tag{A.41}$$

Collect terms in W_g:

$$\left(1 - \beta(1-\lambda) - \frac{\beta^2\lambda\alpha_{wg}}{1 - \beta(1-\alpha_{wg})}\right) W_g = w_g + \frac{\beta\lambda}{1 - \beta(1-\alpha_{wg})}\big(z - (\kappa_g - s_g)\big). \tag{A.42}$$

As before, use the identity

$$\big[1-\beta(1-\lambda)\big]\big[1-\beta(1-\alpha_{wg})\big]-\beta^2\lambda\alpha_{wg}=(1-\beta)\big[1-\beta(1-\alpha_{wg}-\lambda)\big],$$

to obtain

$$W_g=\frac{w_g\big(1-\beta(1-\alpha_{wg})\big)+\beta\lambda\big(z-(\kappa_g-s_g)\big)}{(1-\beta)\big(1-\beta(1-\alpha_{wg}-\lambda)\big)}. \tag{A.43}$$

Step 3: Substituting back into U_g. From (A.36) and (A.43),

$$U_g=\frac{z-(\kappa_g-s_g)}{1-\beta(1-\alpha_{wg})}+\frac{\beta\alpha_{wg}}{1-\beta(1-\alpha_{wg})}\cdot\frac{w_g\big(1-\beta(1-\alpha_{wg})\big)+\beta\lambda\big(z-(\kappa_g-s_g)\big)}{(1-\beta)\big(1-\beta(1-\alpha_{wg}-\lambda)\big)} \tag{A.44}$$

$$=\frac{(1-\beta(1-\lambda))\big(z-(\kappa_g-s_g)\big)+\beta\alpha_{wg}\,w_g}{(1-\beta)\big(1-\beta(1-\alpha_{wg}-\lambda)\big)}. \tag{A.45}$$

Step 4: Worker surplus. Subtract (A.45) from (A.43):

$$W_g-U_g=\frac{w_g-\big(z-(\kappa_g-s_g)\big)}{1-\beta(1-\alpha_{wg}-\lambda)}. \tag{A.46}$$

Step 5: Firm surplus and Nash bargaining. With $V=0$, from (A.32):

$$J_g=(p+\tau_g-w_g)+\beta(1-\lambda)J_g, \tag{A.47}$$

$$J_g\big[1-\beta(1-\lambda)\big]=p+\tau_g-w_g, \tag{A.48}$$

$$J_g=\frac{p+\tau_g-w_g}{1-\beta(1-\lambda)}. \tag{A.49}$$

The Nash condition is

$$(1-\eta)(W_g-U_g)=\eta J_g. \tag{A.50}$$

Substitute surpluses:

$$(1-\eta)\,\frac{w_g-\big(z-(\kappa_g-s_g)\big)}{1-\beta(1-\alpha_{wg}-\lambda)}=\eta\,\frac{p+\tau_g-w_g}{1-\beta(1-\lambda)}. \tag{A.51}$$

Solve for w_g:

$$w_g\left[\frac{1-\eta}{1-\beta(1-\alpha_{wg}-\lambda)}+\frac{\eta}{1-\beta(1-\lambda)}\right]=\frac{(1-\eta)\big(z-(\kappa_g-s_g)\big)}{1-\beta(1-\alpha_{wg}-\lambda)}+\frac{\eta\,(p+\tau_g)}{1-\beta(1-\lambda)}. \tag{A.52}$$

Collecting over the common denominator $1 - \beta(1 - \lambda - \eta\alpha_{wg})$ yields

$$\boxed{w_g = \frac{(1-\eta)\big(z - (\kappa_g - s_g)\big)\,[1 - \beta(1-\lambda)]\ +\ \eta\,(p + \tau_g)\,[1 - \beta(1 - \lambda - \alpha_{wg})]}{1 - \beta\big(1 - \lambda - \eta\alpha_{wg}\big)}}\,. \tag{A.53}$$

Expression (A.53) implies $\partial w_g/\partial\tau_g > 0$, $\partial w_g/\partial s_g > 0$, and $\partial w_g/\partial\kappa_g < 0$, holding tightness (hence α_{wg}) fixed. $\square$